

Linux Essential Commands

1. User and Group Management

`useradd`

Creates a new user account.

Example: `sudo useradd john`

`usermod`

Modifies a user account.

Example: `sudo usermod -aG sudo john`

`userdel`

Deletes a user account.

Example: `sudo userdel john`

`groupadd`

Creates a new user group.

Example: `sudo groupadd developers`

`groupdel`

Deletes a group.

Example: `sudo groupdel developers`

chgrp

Changes the group ownership of a file.

Example: `sudo chgrp developers report.txt`

passwd

Changes a user's password.

Example: `passwd john`

who

Displays who is currently logged in.

Example: `who`

2. File and Directory Management

`ls`

Lists directory contents.

Example: `ls -l /home`

`cd`

Changes the current directory.

Example: `cd /var/log`

`pwd`

Prints the current directory.

Example: `pwd`

`mkdir`

Creates a directory.

Example: `mkdir projects`

`rmdir`

Removes an empty directory.

Example: `rmdir old_folder`

`rm`

Removes files or directories.

Example: `rm -rf temp/`

`cp`

Copies files or directories.

Example: `cp file1.txt backup/`

`mv`

Moves or renames files.

Example: `mv file.txt newname.txt`

`touch`

Creates an empty file or updates timestamp.

Example: `touch index.html`

`find`

Searches for files or directories.

Example: `find /home -name "*.log"`

`locate`

Finds files by name using a database.

Example: `locate nginx.conf`

3. File Content Management

cat

Displays file contents.

Example: `cat notes.txt`

more

Displays file contents page-by-page.

Example: `more largefile.txt`

less

Displays file contents with navigation.

Example: `less largefile.txt`

head

Shows the beginning of a file.

Example: `head -n 10 log.txt`

tail

Shows the end of a file.

Example: `tail -n 10 log.txt`

nano

Opens a file in the Nano text editor.

Example: `nano config.txt`

vi

Opens a file in the Vi editor.

Example: `vi config.txt`

ln

Creates hard or symbolic links.

Example: `ln -s /var/log log_link`

file

Identifies the file type.

Example: `file image.jpg`

stat

Displays file metadata.

Example: `stat notes.txt`

updatedb

Updates the locate database.

Example: `sudo updatedb`

xargs

Builds and executes commands from input.

Example: `find . -name "*.tmp" | xargs rm`

4. File Permissions and Ownership

chmod

Changes file permissions.

Example: `chmod 755 script.sh`

chown

Changes file ownership.

Example: `chown john:admin report.txt`

chgrp

Changes group ownership.

Example: `chgrp developers app.py`

umask

Sets default permissions for new files.

Example: `umask 022`

getfacl

Displays file ACLs.

Example: `getfacl report.txt`

setfacl

Modifies file ACLs.

Example: `setfacl -m u:john:rwx file.txt`

`ls -l`

Lists files with permissions.

Example: `ls -l /etc/passwd`

`stat`

Displays detailed file info.

Example: `stat index.html`

`acl`

General term referring to ACL management using `getfacl` and `setfacl`.

5. Disk Management

df

Displays available disk space.

Example: `df -h`

du

Estimates file/directory size.

Example: `du -sh /var/log`

fdisk

Manages disk partitions.

Example: `sudo fdisk /dev/sda`

parted

Advanced partition tool.

Example: `sudo parted /dev/sdb`

lsblk

Lists block devices.

Example: `lsblk`

blkid

Displays block device attributes.

Example: `sudo blkid`

mount

Mounts a filesystem.

Example: `sudo mount /dev/sdb1 /mnt`

umount

Unmounts a filesystem.

Example: `sudo umount /mnt`

mkfs

Creates a new filesystem.

Example: `sudo mkfs.ext4 /dev/sdb1`

fsck

Checks and repairs filesystems.

Example: `sudo fsck /dev/sda1`

tune2fs

Adjusts filesystem parameters.

Example: `sudo tune2fs -l /dev/sda1`

resize2fs

Resizes a filesystem.

Example: `sudo resize2fs /dev/sda1`

lvcreate

Creates logical volumes (LVM).

Example: `lvcreate -n lv_data -L 10G vg1`

`vgcreate`

Creates volume groups.

Example: `vgcreate vg1 /dev/sdb1`

`pvcreate`

Initializes a physical volume for LVM.

Example: `pvcreate /dev/sdb1`

`lsattr`

Lists file attributes.

Example: `lsattr file.txt`

`chattr`

Changes file attributes.

Example: `chattr +i important.txt`

6. Process Management

`ps`

Shows running processes.

Example: `ps aux`

`top`

Interactive system process monitor.

Example: `top`

`htop`

Advanced interactive process viewer.

Example: `htop`

`kill`

Terminates a process by PID.

Example: `kill 1234`

`killall`

Kills processes by name.

Example: `killall firefox`

`bg`

Resumes suspended job in background.

Example: `bg %1`

fg

Brings background job to foreground.

Example: `fg %1`

nice

Starts process with custom priority.

Example: `nice -n 10 ./script.sh`

renice

Changes running process priority.

Example: `renice -n 5 -p 1234`

jobs

Lists current jobs.

Example: `jobs`

pgrep

Finds process IDs by name.

Example: `pgrep nginx`

pkill

Kills processes by name.

Example: `pkill -f backup.sh`

ps tree

Shows processes in tree format.

Example: `ps tree`

lsof

Lists open files by processes.

Example: `lsof -i :80`

strace

Traces system calls of a process.

Example: `strace ls`

timeout

Runs a command with time limit.

Example: `timeout 5 ping google.com`

watch

Runs a command at intervals.

Example: `watch -n 2 date`

7. Network Management

`ifconfig`

Displays or configures network interfaces.

Example: `ifconfig eth0`

`ip a`

Shows network interfaces and IPs.

Example: `ip a`

`ip r`

Displays routing table.

Example: `ip r`

`ip link set <interface> up/down`

Enables/disables network interface.

Example: `ip link set eth0 down`

`ping`

Sends ICMP echo request.

Example: `ping google.com`

`traceroute`

Shows network path to host.

Example: `traceroute openai.com`

`netstat -tulnp`

Lists ports and services.

Example: `netstat -tulnp`

`ss -tulnp`

Displays socket stats.

Example: `ss -tulnp`

`hostname -I`

Displays IP addresses.

Example: `hostname -I`

`dig`

Queries DNS servers.

Example: `dig example.com`

`nslookup`

Performs DNS lookup.

Example: `nslookup example.com`

`whois`

Displays domain registration info.

Example: `whois example.com`

`curl -I`

Fetches HTTP headers.

Example: `curl -I https://openai.com`

wget

Downloads files.

Example: `wget https://example.com/file.txt`

scp

Securely copies files.

Example: `scp file.txt user@host:/home/user/`

rsync

Efficiently syncs files/directories.

Example: `rsync -avz /src/ user@host:/dest/`

tcpdump

Captures network packets.

Example: `tcpdump -i eth0`

nmap

Scans hosts for open ports.

Example: `nmap 192.168.1.1`

8. System Monitoring

`uptime`

Shows system uptime.

Example: `uptime`

`dmesg`

Displays kernel ring buffer.

Example: `dmesg | less`

`free -m`

Displays memory usage in MB.

Example: `free -m`

`vmstat`

Reports system performance.

Example: `vmstat 1 5`

`iostat`

Shows CPU and I/O usage.

Example: `iostat`

`mpstat`

Displays CPU usage per processor.

Example: `mpstat -P ALL`

`top`

Monitors running processes.

Example: `top`

`htop`

Advanced interactive monitor.

Example: `htop`

`sar -u 5 10`

Monitors CPU usage at intervals.

Example: `sar -u 5 10`

`iostat`

Monitors disk I/O usage.

Example: `iostat`

`lsof`

Lists open files.

Example: `lsof -u john`

`watch -n 2 <command>`

Repeats command every 2 seconds.

Example: `watch -n 2 df -h`

`journalctl -xe`

Shows system logs with extra info.

Example: `journalctl -xe`

`systemctl status`

Checks status of a service.

Example: `systemctl status ssh`

`service status`

Legacy way to check services.

Example: `service apache2 status`

`df -h`

Shows disk usage.

Example: `df -h`

`du -sh`

Shows directory size.

Example: `du -sh /var/log`

`ps aux --sort=-%mem`

Lists processes sorted by memory.

Example: `ps aux --sort=-%mem`

9. Text Processing

`cat`

Displays file contents.

Example: `cat file.txt` — Prints the contents of `file.txt`.

`tac`

Displays file contents in reverse order.

Example: `tac file.txt` — Reverses the order of lines in `file.txt`.

`head`

Shows the first part of a file.

Example: `head -n 5 file.txt` — Displays the first 5 lines.

`tail`

Shows the last part of a file.

Example: `tail -n 5 file.txt` — Displays the last 5 lines.

`tail -f`

Monitors a file for real-time changes.

Example: `tail -f /var/log/syslog` — Follows the `syslog` live.

grep

Searches for patterns in files.

Example: `grep 'error' logfile.txt` — Finds lines with 'error'.

grep -r

Recursively searches directories.

Example: `grep -r 'TODO' /project` — Finds all TODO comments.

awk

Pattern scanning and processing language.

Example: `awk '{print $1, $3}' file.txt` — Prints first and third fields.

sed

Stream editor for filtering text.

Example: `sed 's/foo/bar/g' file.txt` — Replaces 'foo' with 'bar'.

cut

Cuts sections from each line.

Example: `cut -d ':' -f1 /etc/passwd` — Displays only usernames.

sort

Sorts lines in text files.

Example: `sort names.txt` — Alphabetically sorts file.

uniq

Filters duplicate lines.

Example: `sort names.txt | uniq` — Removes duplicates.

wc -l

Counts lines in a file.

Example: `wc -l file.txt` — Outputs number of lines.

wc -w

Counts words in a file.

Example: `wc -w file.txt` — Outputs word count.

wc -c

Counts bytes in a file.

Example: `wc -c file.txt` — Outputs file size in bytes.

tr

Translates or deletes characters.

Example: `tr 'a-z' 'A-Z' < file.txt` — Converts lowercase to uppercase.

diff

Compares two files line by line.

Example: `diff file1.txt file2.txt` — Shows differences.

cmp

Compares files byte by byte.

Example: `cmp file1.bin file2.bin` — Returns first mismatch.

tee

Saves output to file and displays it.

Example: `echo 'Log entry' | tee log.txt` — Outputs and writes to file.

10. Package Management

For Debian-based systems:

`apt update`

Updates package lists.

Example: `sudo apt update` — Refreshes available packages.

`apt upgrade`

Upgrades all installed packages.

Example: `sudo apt upgrade` — Installs latest versions.

`apt install`

Installs a package.

Example: `sudo apt install curl` — Installs curl.

`apt remove`

Removes a package.

Example: `sudo apt remove nano` — Uninstalls nano.

`apt purge`

Removes package and config files.

Example: `sudo apt purge apache2` — Cleans Apache fully.

`dpkg -i`

Installs .deb files manually.

Example: `sudo dpkg -i package.deb` — Installs local package.

`dpkg -r`

Removes a package.

Example: `sudo dpkg -r package` — Deletes package.

`dpkg -l`

Lists all installed packages.

Example: `dpkg -l` — Displays installed package list.

`apt-cache search`

Searches for packages.

Example: `apt-cache search nginx` — Finds nginx packages.

`apt autoremove`

Cleans up unused dependencies.

Example: `sudo apt autoremove` — Frees disk space.

`apt clean`

Cleans package cache.

Example: `sudo apt clean` — Removes cached .deb files.

For RHEL-based systems:

`yum update / dnf update`

Updates installed packages.

Example: `sudo dnf update` — Updates all packages.

`yum install / dnf install`

Installs new packages.

Example: `sudo dnf install git` — Installs Git.

`yum remove / dnf remove`

Removes installed packages.

Example: `sudo yum remove vim` — Uninstalls vim.

`rpm -ivh`

Installs .rpm package.

Example: `sudo rpm -ivh file.rpm` — Installs a local .rpm file.

`rpm -e`

Erases a package.

Example: `sudo rpm -e package` — Removes an rpm package.

`yum list installed / dnf list installed`

Lists all packages.

Example: `dnf list installed` — Shows installed software.

```
yum search / dnf search
```

Searches the repository.

Example: `yum search python` — Finds Python-related packages.

```
yum clean all
```

Cleans all cached data.

Example: `sudo yum clean all` — Frees up space.

11. Archive and Compression

`tar -cvf`

Creates a tar archive.

Example: `tar -cvf backup.tar folder/` — Archives folder.

`tar -xvf`

Extracts a tar archive.

Example: `tar -xvf backup.tar` — Extracts contents.

`tar -tvf`

Lists contents of a tar archive.

Example: `tar -tvf backup.tar` — Views file list.

`tar -czvf`

Creates a compressed archive.

Example: `tar -czvf backup.tar.gz folder/` — Compresses to .tar.gz.

`tar -xzvf`

Extracts compressed archive.

Example: `tar -xzvf backup.tar.gz` — Unpacks gzip tarball.

`tar -cJvf`

Creates a tar.xz archive.

Example: `tar -cJvf archive.tar.xz folder/` —
Compresses to .xz format.

`tar -xJvf`

Extracts tar.xz archive.

Example: `tar -xJvf archive.tar.xz` — Unpacks xz
archive.

`zip`

Creates a zip archive.

Example: `zip archive.zip file1 file2` —
Compresses files.

`unzip`

Extracts files from zip archive.

Example: `unzip archive.zip` — Extracts zip contents.

`unzip -l`

Lists contents of a zip.

Example: `unzip -l archive.zip` — Views inside
archive.

`zip -r`

Recursively zips a directory.

Example: `zip -r project.zip project/` — Zips
entire folder.

gzip

Compresses a file.

Example: `gzip log.txt` — Creates log.txt.gz.

gunzip

Decompresses a gzip file.

Example: `gunzip log.txt.gz` — Restores original file.

bzip2

Compresses with bzip2.

Example: `bzip2 bigfile` — Makes bigfile.bz2.

bunzip2

Decompresses bzip2 file.

Example: `bunzip2 bigfile.bz2` — Restores bigfile.

12. Shutdown and Reboot

`shutdown -h now`

Shuts down the system immediately.

Example: `sudo shutdown -h now`

`shutdown -h +10`

Schedules shutdown after 10 minutes.

Example: `sudo shutdown -h +10`

`shutdown -h 22:00`

Schedules shutdown at a specific time.

Example: `sudo shutdown -h 22:00`

`shutdown -c`

Cancels a scheduled shutdown.

Example: `sudo shutdown -c`

`poweroff`

Powers off the machine.

Example: `sudo poweroff`

`halt`

Stops all processes and halts.

Example: `sudo halt`

`reboot`

Reboots the system.

Example: `sudo reboot`

`shutdown -r now`

Immediate reboot.

Example: `sudo shutdown -r now`

`shutdown -r +5`

Reboot in 5 minutes.

Example: `sudo shutdown -r +5`

`systemctl reboot`

Systemd reboot command.

Example: `sudo systemctl reboot`

13. Task Scheduling

`crontab -e`

Edits cron jobs.

Example: `crontab -e` — Adds a scheduled task.

`crontab -l`

Lists scheduled tasks.

Example: `crontab -l` — Displays your cron jobs.

`crontab -r`

Removes all cron jobs.

Example: `crontab -r` — Deletes your entire crontab.

`crontab -u <user> -l`

Lists cron jobs for a specific user.

Example: `crontab -u john -l`

`at`

Schedules a one-time task.

Example: `echo "shutdown -h now" | at 10:30`

14. System Diagnostics and Troubleshooting

`journalctl -xe`

Views detailed logs.

Example: `journalctl -xe` — Useful for errors.

`journalctl -f`

Follows logs live.

Example: `journalctl -f` — Similar to `tail -f`.

`journalctl --since "1 hour ago"`

Shows logs from the past hour.

Example: `journalctl --since "1 hour ago"`

`dmesg | less`

Views boot and hardware logs.

Example: `dmesg | less`

`dmesg | grep error`

Filters kernel logs for errors.

Example: `dmesg | grep error`

`cat /var/log/syslog`

Displays general system log (Debian).

Example: `cat /var/log/syslog`

`cat /var/log/messages`

Displays system logs (RHEL).

Example: `cat /var/log/messages`

`uptime`

Shows system uptime and load.

Example: `uptime`

`free -m`

Displays memory usage in MB.

Example: `free -m`

`top`

Real-time process and usage monitor.

Example: `top`

`htop`

Improved real-time monitor.

Example: `htop`

`vmstat`

System performance metrics.

Example: `vmstat 1 5`

`iostat`

CPU and disk usage.

Example: `iostat`